

Lab: Variational Inference

Learning Goal: Build intuition for approximate inference through worked examples and conceptual exercises.

Part 1: Why Exact Inference Fails

Exercise: Consider a simple model with 3 binary hidden states (s_1, s_2, s_3).

1. List all possible state combinations (there should be $2^3 = 8$).
2. Now consider 10 binary states: How many combinations? 20 states? 50 states?
3. If evaluating each combination takes 1 nanosecond, how long would it take to evaluate all combinations for 50 binary states?
4. Why does this demonstrate the need for approximate inference?

Write your response here

Part 2: KL Divergence Computation

Learning Goal: Compute and interpret KL divergence for simple distributions.

Exercise: Consider two discrete distributions over outcomes $\{A, B, C\}$:

- $P = \{0.5, 0.3, 0.2\}$
- $Q_1 = \{0.4, 0.35, 0.25\}$
- $Q_2 = \{0.1, 0.1, 0.8\}$
- Compute $D_{KL}[Q_1 \parallel P] = \sum Q_1(x) \times \log(Q_1(x)/P(x))$
- Compute $D_{KL}[Q_2 \parallel P]$
- Which approximation (Q_1 or Q_2) is closer to P ? Does this match your intuition?

Write your response here

Part 3: The ELBO Decomposition

Learning Goal: Understand accuracy vs. complexity in model fitting.

Scenario: Two students build models to predict exam scores:

- Student A's model: "Everyone gets a B" (simple but inaccurate)
- Student B's model: "Score depends on study hours, GPA, sleep, mood, day of week, horoscope, weather, shirt color..." (accurate on training data but overly complex)
- Which model has higher accuracy? Which has lower complexity?
- Which model would the ELBO prefer? Why?
- What would an optimal model look like?

Write your response here

Part 4: Connecting Free Energy to Cognition

Learning Goal: Map mathematical quantities to cognitive concepts.

Mathematical Quantity	Cognitive Equivalent
$q(s)$	
$P(s o)$	
F (free energy)	
$D_{KL}[q p]$	
Accuracy term	
Complexity term	

Fill in the cognitive equivalents (hint: beliefs, perfect inference, surprise, error, fit, simplicity).

Write your response here

Part 5: Reflection

In 150 words, reflect: The brain can't perform exact Bayesian inference, so it approximates. Does this mean our beliefs are always "wrong" to some degree? Is approximate inference a limitation or an advantage?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Exponential growth analysis	Intractability of exact inference
2	KL divergence computation	Measuring distribution closeness
3	Trade-off analysis	ELBO accuracy-complexity balance
4	Concept mapping	Math $\leftrightarrow$ Cognition bridge
5	Epistemological reflection	Approximate vs. exact inference